

MIP_MASTER_ROADMAP.md

Mainframe Intelligence Platform

End-to-End Delivery Roadmap

Version 1.0

Vision

Transform legacy application source code into enterprise intelligence.

Phase 0

Foundation

Goal:

Create engineering platform.

Deliverables:

- VSCode Workspace
- Repository Structure
- SQLite
- NetworkX
- Google ADK Setup
- CI/CD
- Unit Test Framework

Output:

Working engineering environment.

Success Criteria:

Developer can run MIP locally.

Phase 1

Discovery

Goal:

Know what exists.

Deliverables:

- Repository Scanner
- Artifact Classifier
- Program Catalog
- Job Catalog

Questions Answered:

What programs exist?

What jobs exist?

What copybooks exist?

Output:

Application Inventory

Phase 2

Call Graph Discovery

Goal:

Understand program relationships.

Deliverables:

- CALL Parser
- CALL Relationship Store
- Call Graph

Questions Answered:

Who calls program X?

What does program X call?

Output:

Application Dependency Graph

Phase 3

Copybook Intelligence

Goal:

Understand shared structures.

Deliverables:

- COPY Parser
- Copybook Catalog
- Program-Copybook Relationships

Questions Answered:

Which programs use copybook X?

What copybooks does program Y use?

Output:

Shared Structure Graph

Phase 4

Data Lineage

Goal:

Understand data usage.

Deliverables:

- EXEC SQL Parser
- DB2 Metadata
- Table Relationships

Questions Answered:

Which tables are updated?

Which programs read a table?

Output:

Data Lineage Model

Phase 5

Knowledge Graph

Goal:

Create unified graph.

Deliverables:

- NetworkX Graph Builder
- Graph Persistence
- Graph Query Layer

Questions Answered:

Show complete dependency chain.

Output:

Enterprise Knowledge Graph

Phase 6

Reasoning Engine

Goal:

Move beyond discovery.

Deliverables:

- Impact Analysis
- Root Program Discovery
- Flow Discovery

Questions Answered:

What breaks if this changes?

Output:

Enterprise Reasoning Layer

Phase 7

Google ADK Copilot

Goal:

Natural language interaction.

Deliverables:

- ADK Agent
- Graph Tools
- Metadata Tools

Questions Answered:

Explain payment processing.

Output:

Legacy Intelligence Copilot

Phase 8

Business Rule Discovery

Goal:

Discover hidden business logic.

Deliverables:

- Rule Extractor
- Rule Catalog
- Capability Mapping

Questions Answered:

What business rules exist?

Output:

Business Knowledge Layer

Phase 9

Modernization Intelligence

Goal:

Guide transformation.

Deliverables:

- Service Discovery
- API Discovery
- Event Discovery
- Migration Recommendations

Questions Answered:

How should this system be modernized?

Output:

Modernization Advisor

Final Outcome

Source Code ↓ Metadata ↓ Relationships ↓ Knowledge Graph ↓ Reasoning ↓ Copilot ↓
Modernization Intelligence

MIP becomes the Enterprise Legacy Intelligence Platform.